

Figure S1. Body length of beetle species per feeding group in A: pine, B: new recovering fynbos, C: old recovering fynbos and D: fynbos

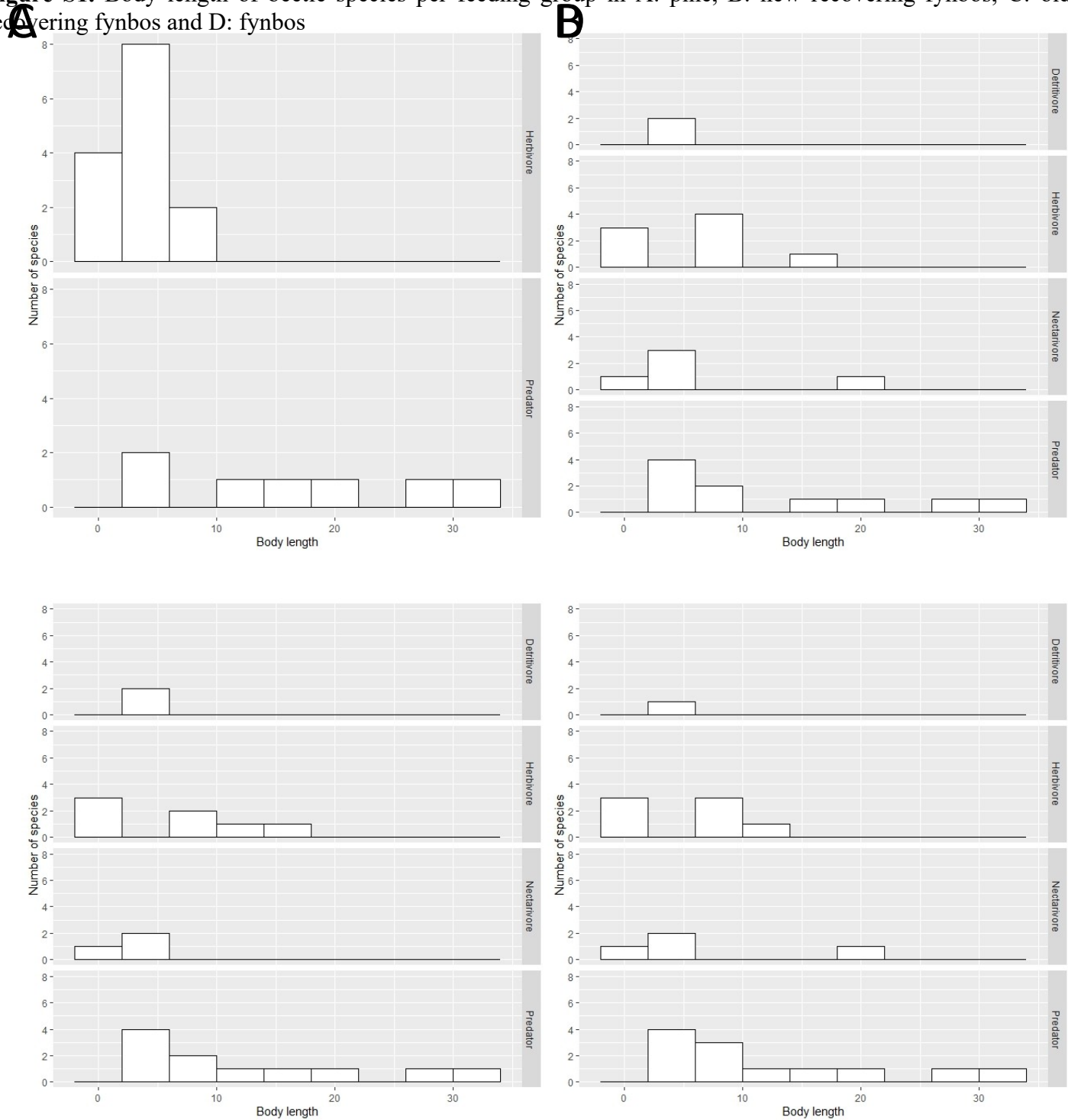
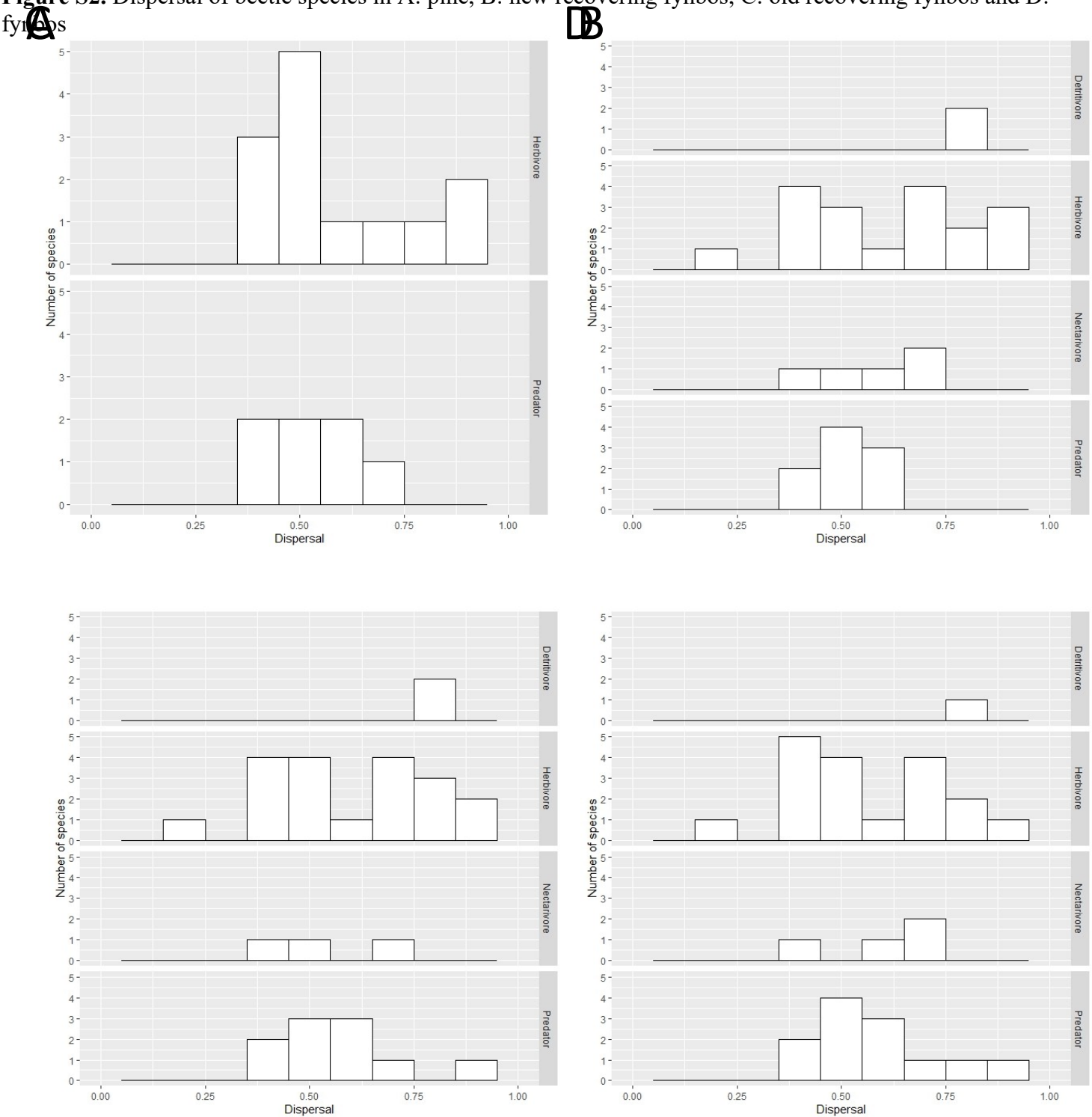
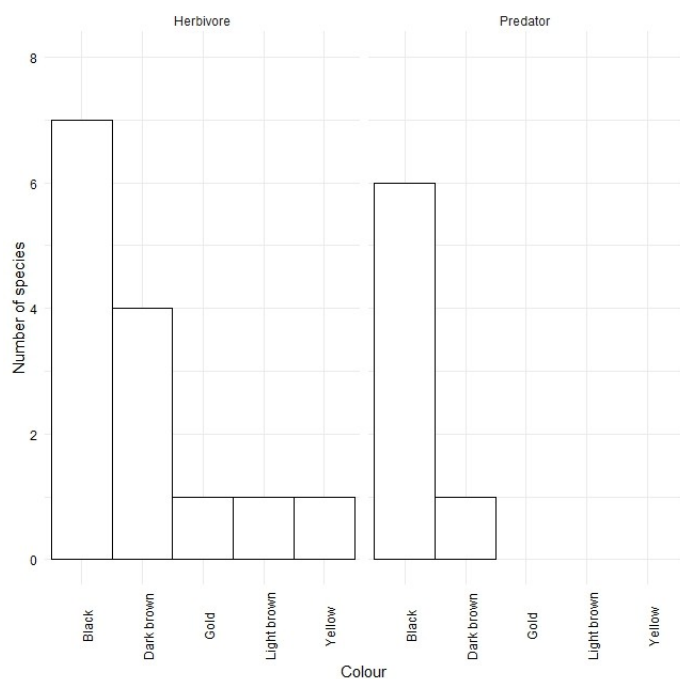


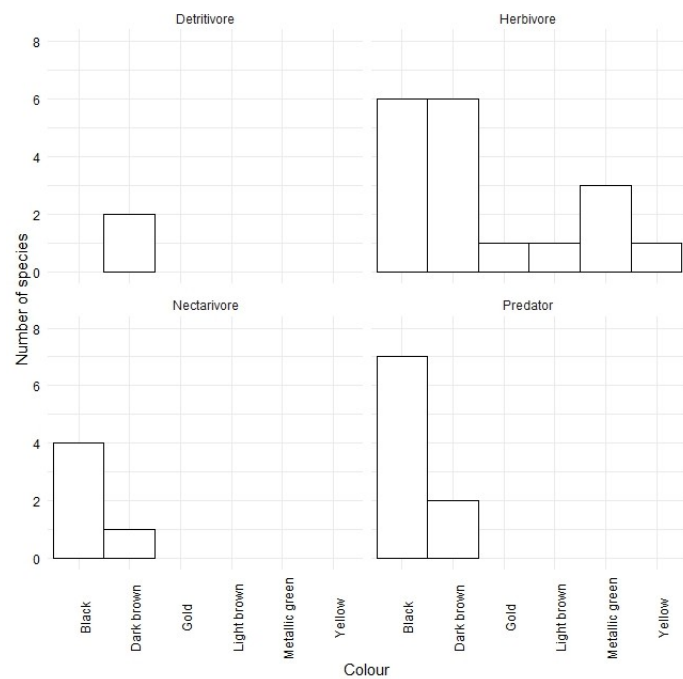
Figure S2. Dispersal of beetle species in A: pine, B: new recovering fynbos, C: old recovering fynbos and D: fynbos



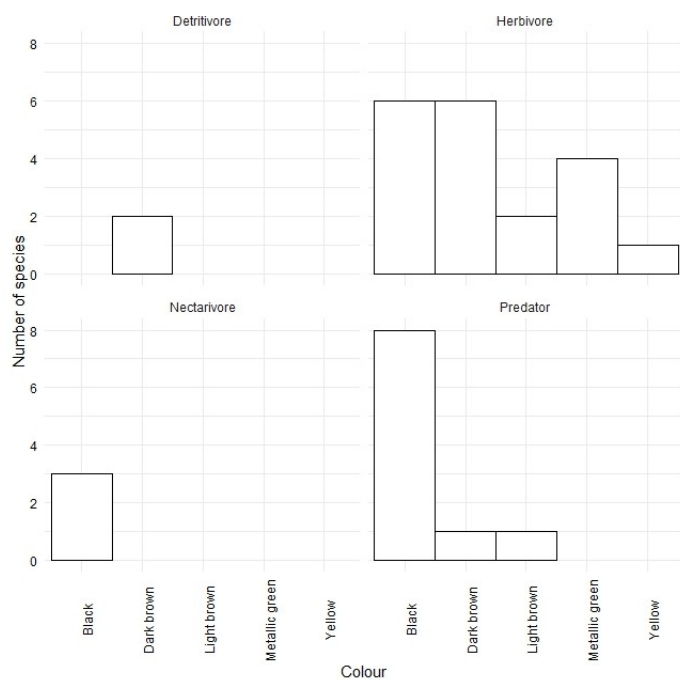
A



B



C



D

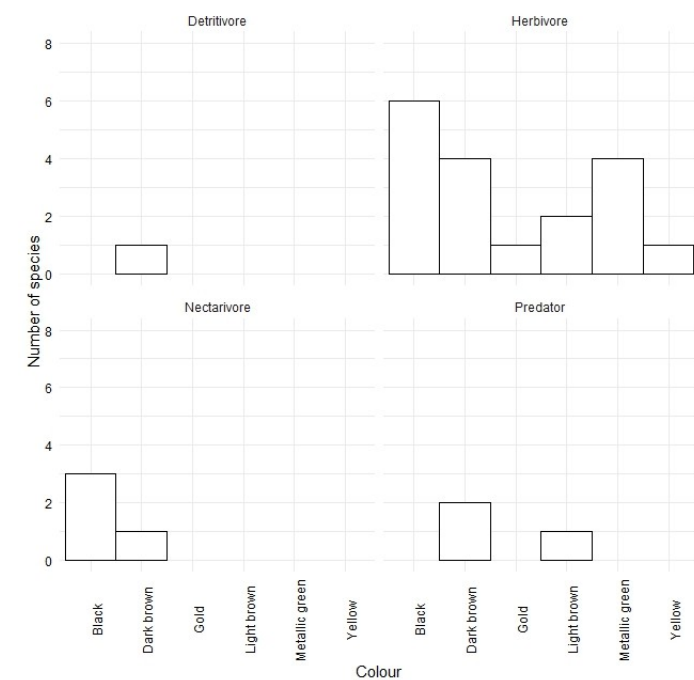


Figure S3. Colour of beetle species in A: pine, B: new recovering fynbos, C: old recovering fynbos and D: fynbos

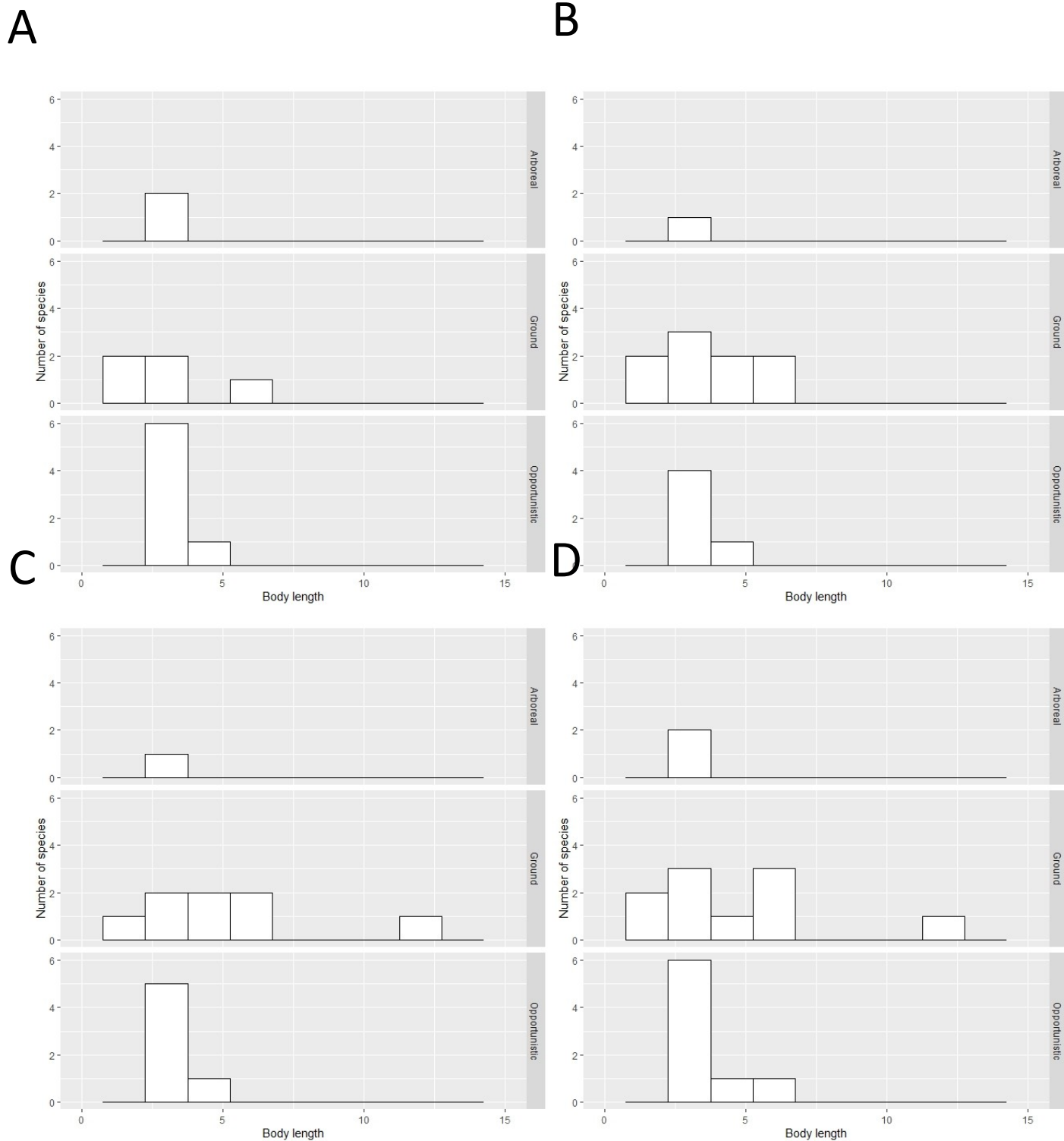
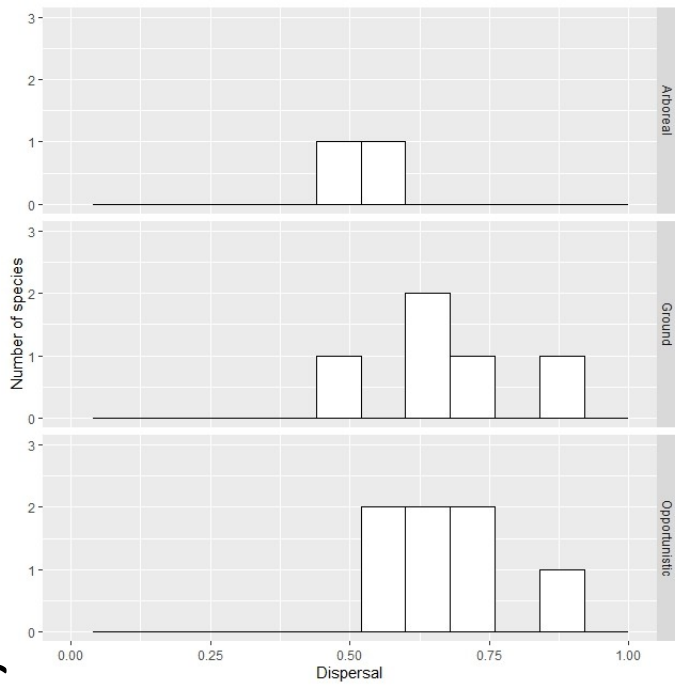
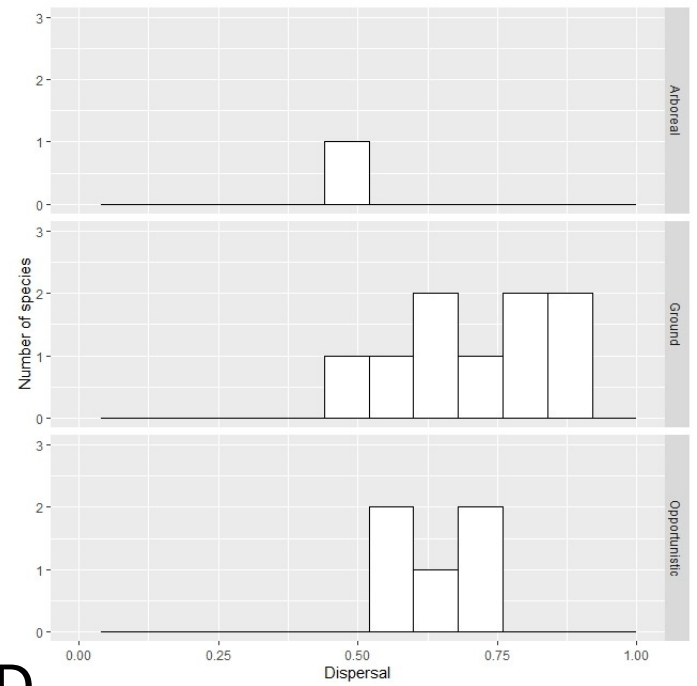


Figure S4. Body length of ant species per nesting type in A: pine, B: new recovering fynbos, C: old recovering fynbos and D: fynbos

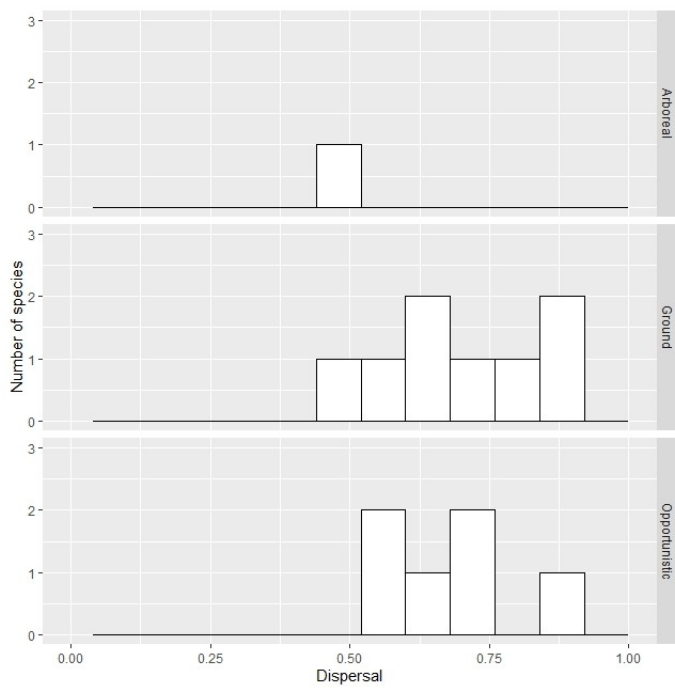
A



B



C



D

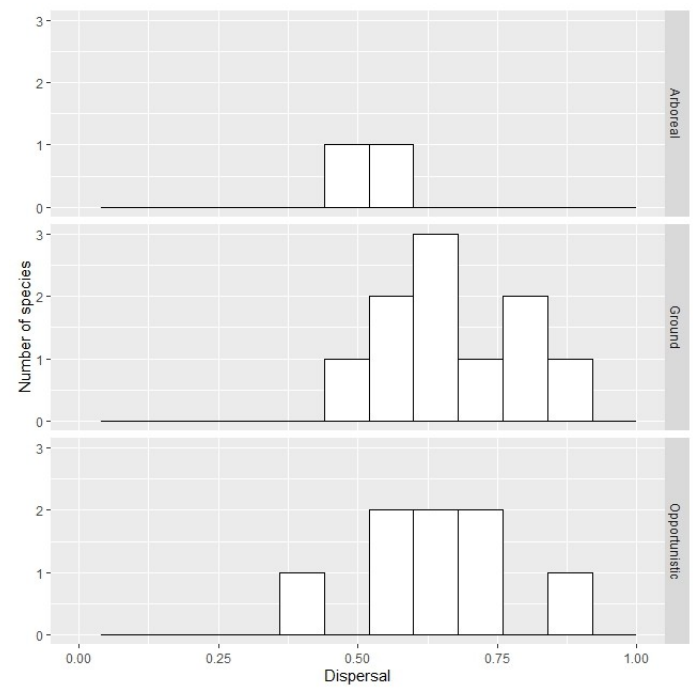
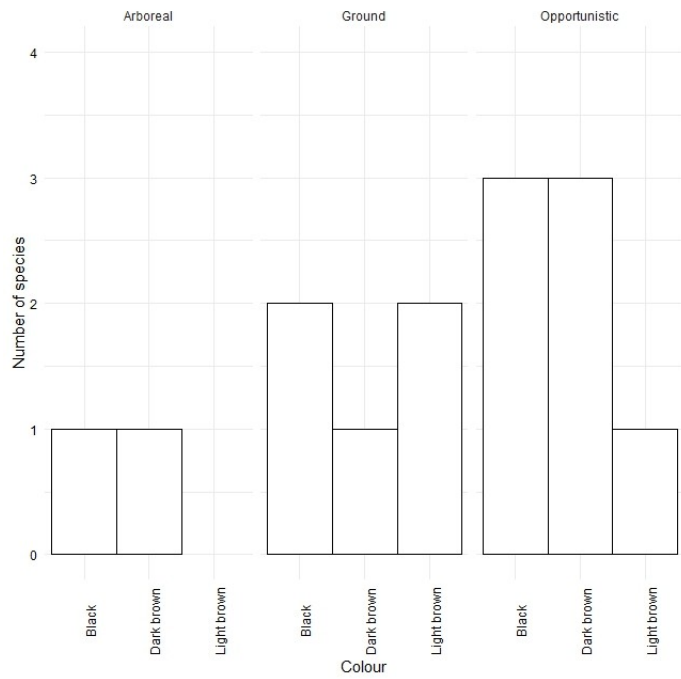
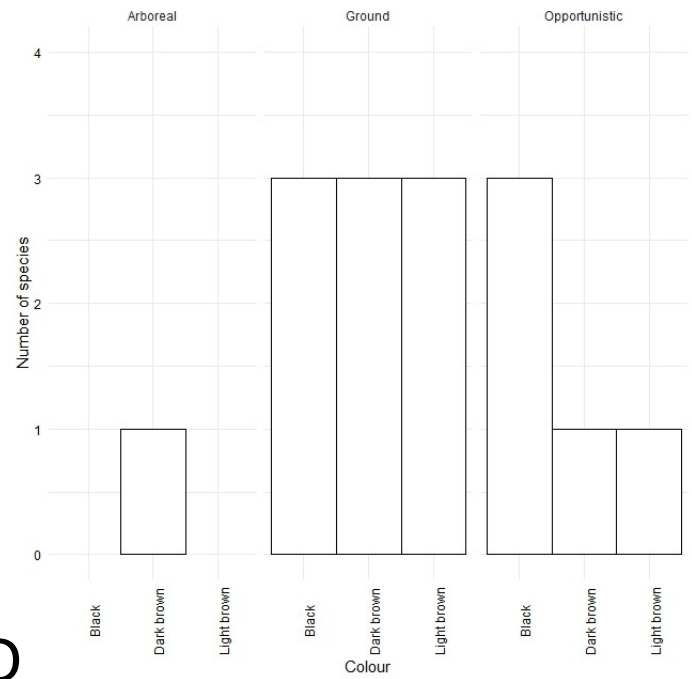


Figure S5. Dispersal of ant species per nesting type in A: pine, B: new recovering, C: old recovering and D: fynbos

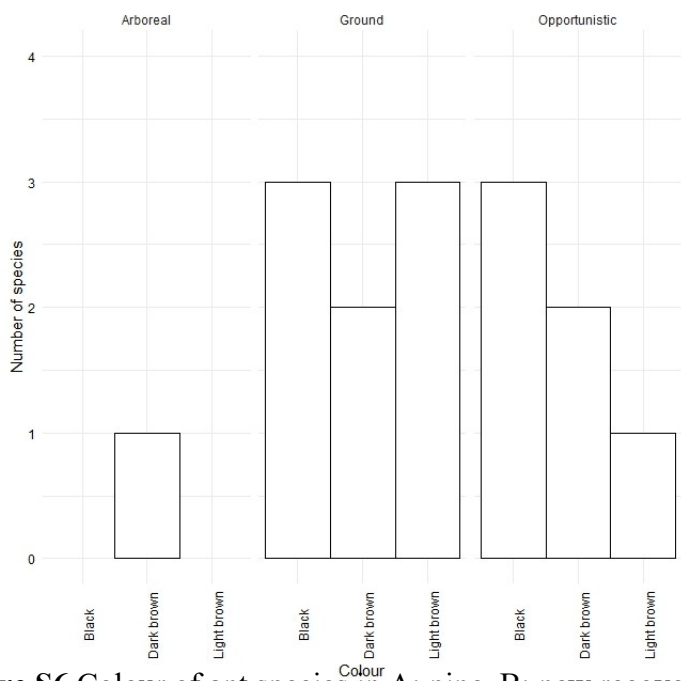
A



B



C



D

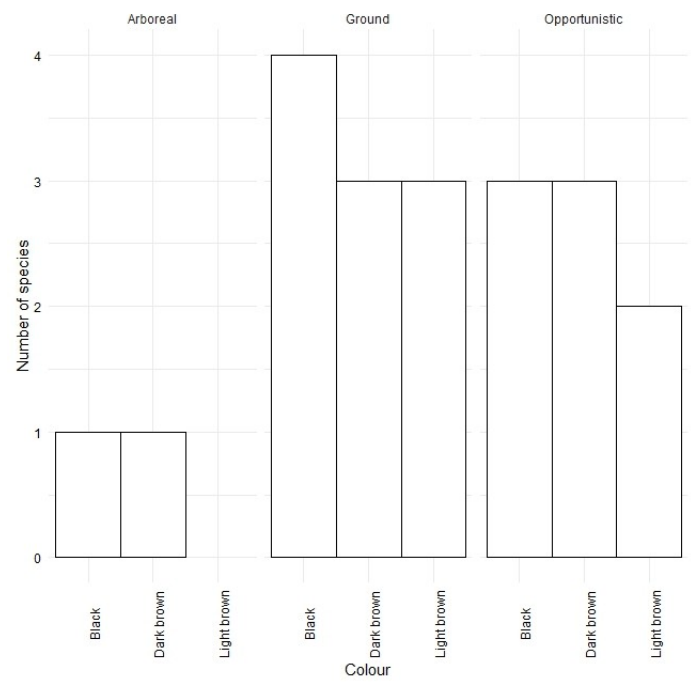
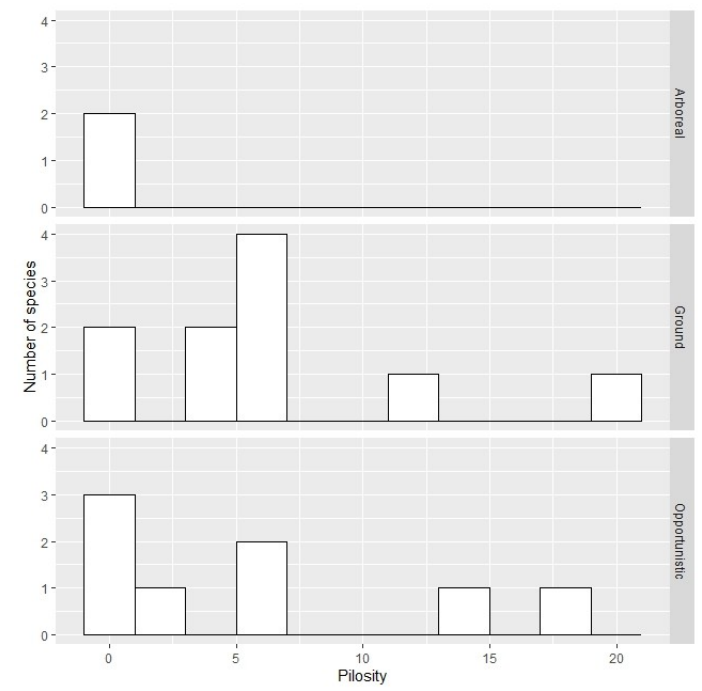
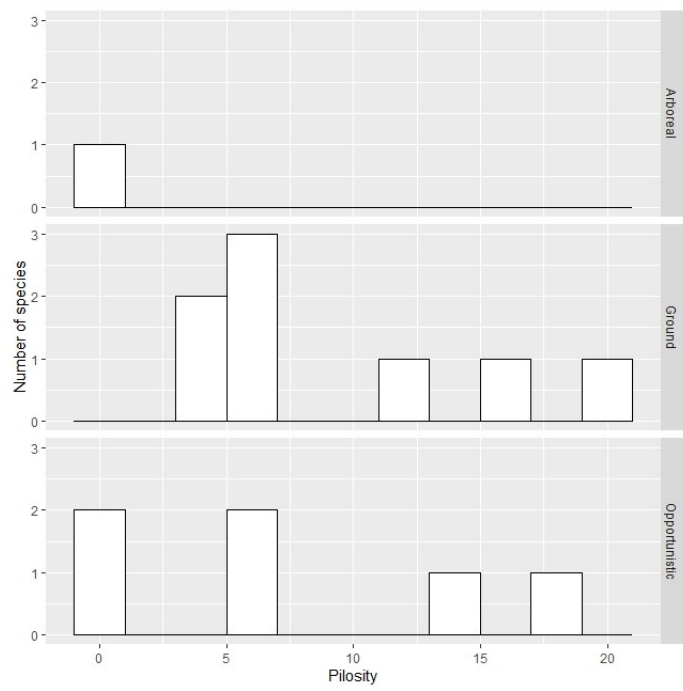
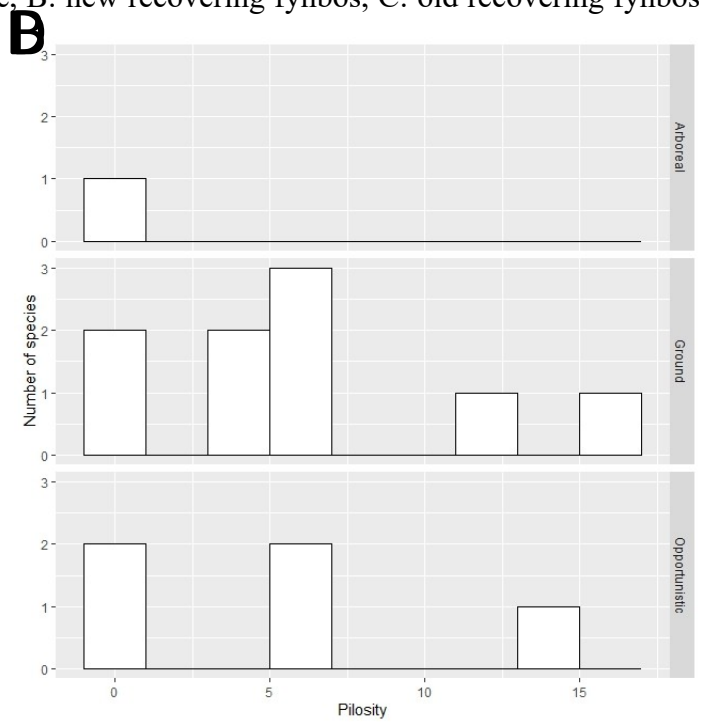
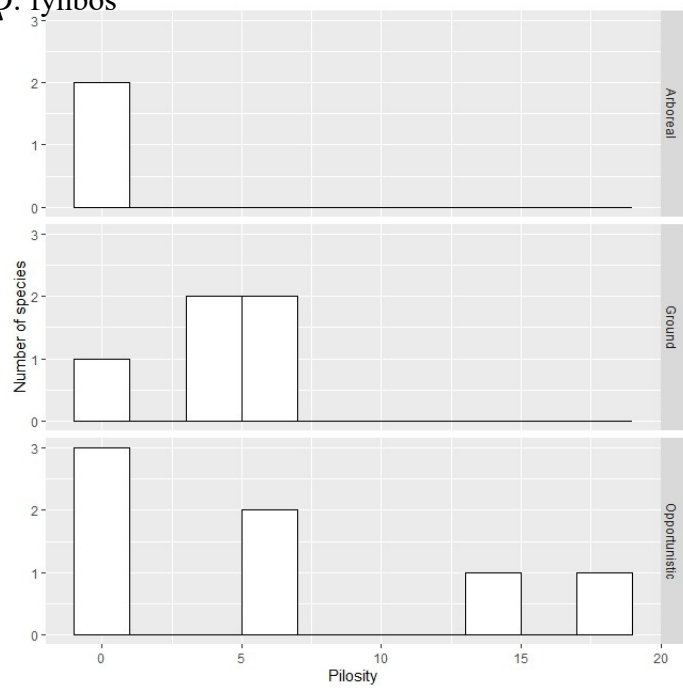
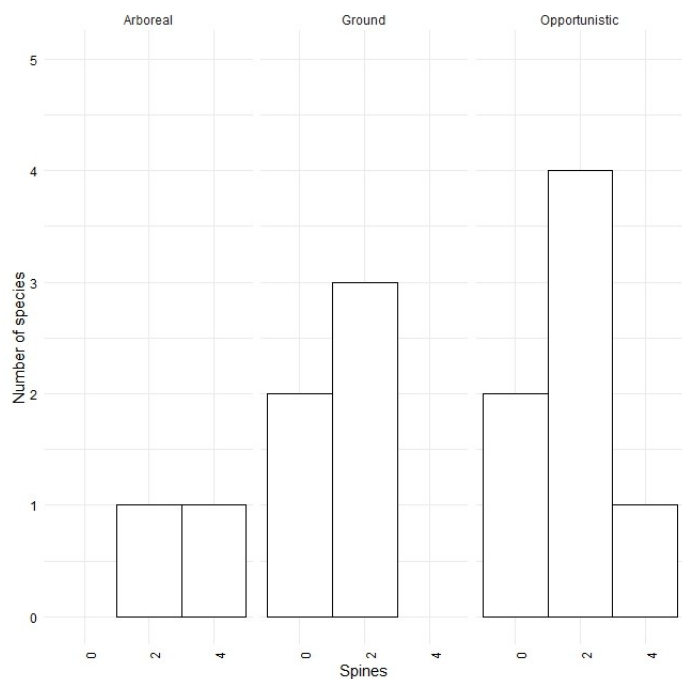


Figure S6 Colour of ant species in A: pine, B: new recovering, C: old recovering and D: fynbos

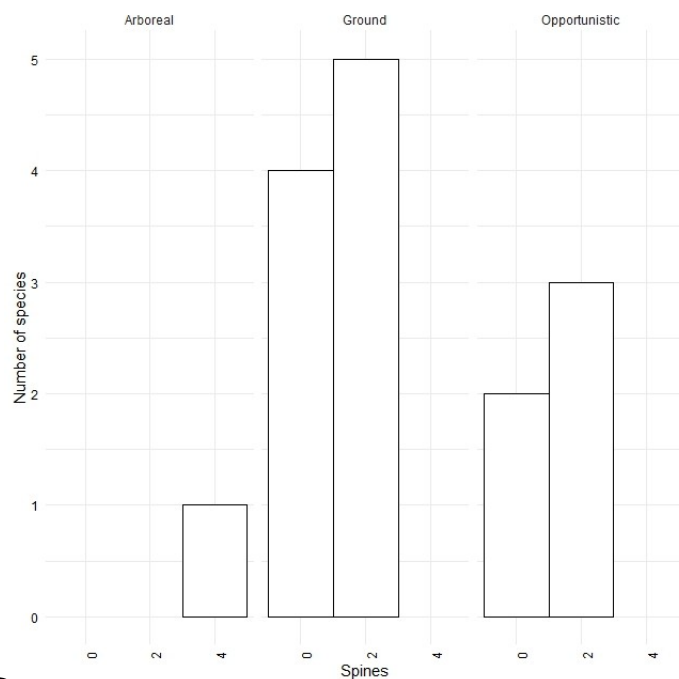
Figure S7. Pilosity of ant species per nesting type in A: pine, B: new recovering fynbos, C: old recovering fynbos and D: fynbos



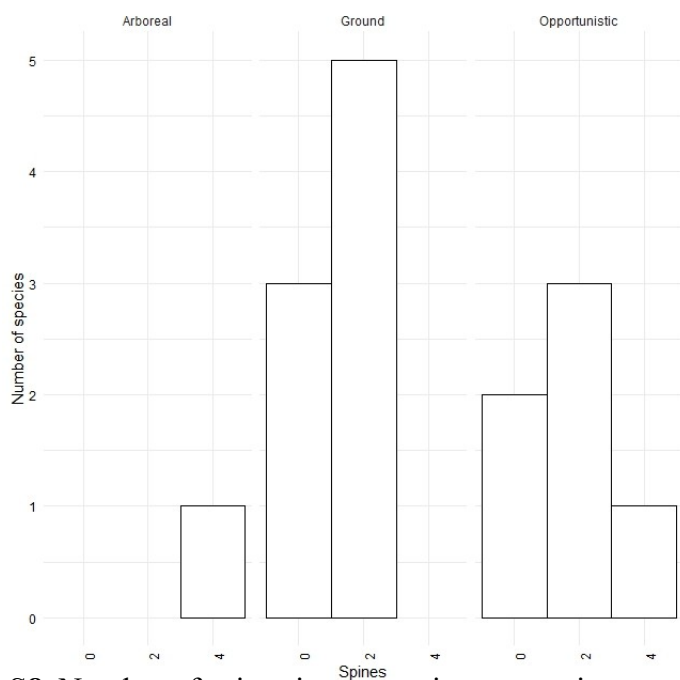
A



B



C



D

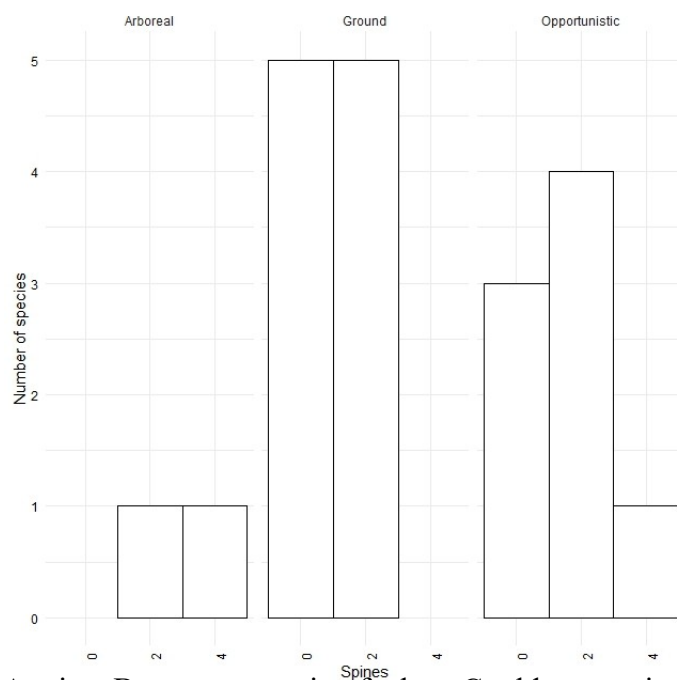


Figure S8. Number of spines in ant species per nesting type in A: pine, B: new recovering fynbos, C: old recovering fynbos and D: fynbos

Table S1. Beetle morphospecies codes and the corresponding species traits.

Spp code	Family	Feeding guild	Body length (mm)	Dispersal	Colour
Col003	Mordellidae	Nectarivore	4.482	0.352	Black
Col004	Scarabaeidae	Herbivore	9.85	0.507	Dark brown
Col005	Melyridae	Herbivore	5.387	0.247	Metallic green
Col007	Chrysomelidae	Herbivore	2.005	0.4	Light brown
Col008	Staphylinidae	Predator	3.292	0.475	Dark brown
Col009	Carabidae	Predator	30.671	0.387	Black
Col010	Carabidae	Predator	16.932	0.586	Black
Col012	Carabidae	Predator	12.009	0.706	Black
Col013	Curculionidae	Herbivore	5.153	0.528	Dark brown
Col015	Staphylinidae	Predator	28.065	0.445	Black
Col018	Chrysomelidae	Herbivore	2.206	0.423	Black
Col023	Scarabaeidae	Detritivore	2.381	0.757	Dark brown
Col027	Carabidae	Predator	18.846	0.545	Black
Col029	Scarabaeidae	Detritivore	4.289	0.838	Dark brown
Col030	Curculionidae	Herbivore	2.107	0.733	Dark brown
Col041	Carabidae	Predator	3.261	0.623	Black
Col042	Chrysomelidae	Herbivore	1.764	0.417	Black
Col050	Scarabaeidae	Herbivore	15.892	0.726	Dark brown
Col054	Chrysomelidae	Herbivore	2.015	0.866	Light brown
Col060	Curculionidae	Herbivore	1.566	0.508	Black
Col066	Curculionidae	Herbivore	5.992	0.844	Black
Col068	Chrysomelidae	Herbivore	1.376	0.462	Black
Col074	Chrysomelidae	Herbivore	1.838	0.375	Black
Col093	Chrysomelidae	Herbivore	4.667	0.662	Gold
Col094	Mordellidae	Nectivore	3.603	0.463	Black
Col098	Chrysomelidae	Herbivore	4.519	0.482	Yellow
Col101	Melyridae	Herbivore	9.408	0.378	Dark brown
Col104	Tenebrionidae	Predator	6.264	0.51	Dark brown
Col106	Mordellidae	Nectarivore	1.96	0.673	Black
Col108	Scarabaeidae	Herbivore	5.992	0.808	Dark brown
Col109	Meloidae	Herbivore	13.396	0.707	Metallic green
Col115	Chrysomelidae	Herbivore	6.481	0.68	Metallic green
Col116	Scarabaeidae	Nectivore	19.726	0.75	Dark brown
Col119	Chrysomelidae	Herbivore	4.262	0.803	Metallic green
Col123	Carabidae	Predator	15.71	0.783	Black
Col130	Melyridae	Herbivore	6.535	0.352	Black
Col132	Carabidae	Predator	9.052	0.539	Black
Col133	Carabidae	Predator	4.862	0.6	Black
Col136	Carabidae	Predator	8.934	0.856	Light brown
Col137	Chrysomelidae	Herbivore	2.855	0.583	Black
Col142	Scarabaeidae	Detritivore	4.125	0.772	Dark brown
Col154	Curculionidae	Herbivore	5.362	0.889	Black
Col172	Scarabaeidae	Herbivore	5.248	0.915	Dark brown

Col176	Curculionidae	Herbivore	4.449	0.649	Black
Col178	Dermestidae	Nectarivore	2.322	0.57	Black

Table S2. Ant species codes and their corresponding trait values

Spp Code	Scientific Name	Body length (mm)	Dispersal	Colour	Nesting type	No. of spines	Pilosity	Sculpturing
For003	<i>Lepisiota capensis</i>	2.675	0.758	Black Light	Ground	2	6	2
For006	<i>Monomorium tablense</i>	1.802	0.497	brown	Ground	2	5	1
For008	<i>Camponotus werthi</i>	2.447	0.714	Black	Opportunistic	2	7	1
For009	<i>Crematogaster capensis</i>	3.226	0.46	Dark brown	Arboreal	4	0	2
For011	<i>Tetramorium</i>	3.615	0.635	Light brown	Opportunistic	2	6	1
For015	<i>Tetramorium sericeiventre</i>	2.824	0.647	Dark brown	Opportunistic	2	0	2
For016	<i>Crematogaster delagoensis</i>	3.608	0.524	Black	Arboreal	2	0	2
For017	<i>Technomyrmex pallipes</i>	2.865	0.597	Black	Opportunistic	0	0	2
For018	<i>Technomyrmex</i>	2.907	0.582	Dark brown	Opportunistic	0	0	2
For019	<i>Camponotus bianconii</i>	6.041	0.863	Dark brown	Ground	0	4	1
For021	<i>Monomorium boerorum</i>	2.015	0.668	Black	Ground	0	0	1
For024	<i>Tetramorium capense</i>	3.136	0.878	Dark brown	Opportunistic	4	18	2
For026	<i>Myrmicaria nigra</i>	6.341	0.768	Black	Ground	2	12	2
For027	<i>Monomorium havilandi</i>	3.08	0.835	Dark brown	Ground	0	0	1
For030	<i>Pheidole tenuinodis</i>	2.839	0.671	Light brown	Ground	2	6	2
For033	<i>Pheidole caffra</i>	4.576	0.587	Light brown	Ground	2	7	1
For035	<i>Tetramorium grassii</i>	4.146	0.721	Black	Opportunistic	2	15	2
For036	<i>Bothroponera cavernosa</i>	12.171	0.624	Black	Ground	0	20	2
For037	<i>Messor denticornis</i>	5.036	0.845	Dark brown	Ground	0	16	1
For042	<i>Leptogenys castanea</i>	6.021	0.382	Light brown	Opportunistic	0	2	1
For048	<i>Camponotus</i> sp.	6.21	0.556	Dark brown	Ground	0	6	1

